ModelIt!

Teacher's Guide

Where Does Rain Come From?

The Amazing Water Cycle

Quick Reference

Grade Level	Kindergarten
NGSS Standard	K-ESS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: K-ESS2-1

Use and share observations of local weather conditions to describe patterns over time.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students create a simple bag model of the water cycle, observing evaporation and condensation in real time.
Disciplinary Core Idea	**ESS2.D Weather and Climate** — Weather is the combination of sunlight, wind, snow or rain, and temperature. Water cycles between the ground and the atmosphere through evaporation and precipitation.
Crosscutting Concept	**Cause and Effect** — Students observe that the sun's heat (cause) makes water evaporate faster (effect), driving the water cycle.

CAST Testing Alignment

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

The water cycle is one of nature's most important processes. Every drop of rain that falls has made the journey from Earth's surface to the atmosphere and back countless times over billions of years.

How the Water Cycle Works

The water cycle has three main stages. Evaporation occurs when the sun heats water on Earth's surface (oceans, lakes, rivers, puddles), turning liquid water into water vapor — an invisible gas that rises into the atmosphere. Condensation happens when water vapor rises high enough to encounter cold air, causing it to turn back into tiny liquid droplets that cling together to form clouds. Precipitation occurs when cloud droplets combine and grow heavy enough to fall as rain, snow, sleet, or hail. The water then collects in bodies of water and the cycle repeats.

Why the Sun Drives the Cycle

The sun is the engine of the water cycle. Solar energy heats water at Earth's surface, providing the energy needed for evaporation. Hotter temperatures cause faster evaporation — a puddle on a hot summer day disappears much faster than one on a cool cloudy day. Without the sun's heat, the water cycle would stop and there would be no rain, no rivers, and no fresh water. Understanding that the sun's heat controls evaporation speed is the key concept for kindergarteners.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Sun's Heat as the external component (the sun's warmth is controlled by weather/seasons) and Evaporation Speed as the internal component (it changes based on how warm the sun makes the water).

E **Establish Relationships**
Students discover that more Sun's Heat leads to faster Evaporation Speed — a positive relationship.

V **Visualize & Model**
Students build a simple two-part model connecting Sun's Heat to Evaporation Speed with a positive arrow.

E **Evaluate Outcomes**
Students test Cool Day vs. Hot Day scenarios and observe how quickly puddles dry up or water rises in a bag.

R **Revise & Extend**
Students explore what happens when they add wind — does wind make puddles dry up even faster?

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Have you ever seen a puddle disappear? Where did the water go? And where does rain come from? Today we solve BOTH mysteries!"*

Watch For

Show a video or time-lapse of a puddle drying up. Ask: The water disappeared! Did someone drink it? Where did it go?

Slide 2: What Will We Learn?

3 min

Say: *"Today we learn about the WATER CYCLE — the amazing journey water takes from the ground to the sky and back! The water in your drink has been on this journey millions of times!"*

Watch For

Introduce 'evaporation' and 'water cycle.' For evaporation, wiggle fingers upward like water floating into the sky. Practice the word together.

Slide 3: Let's Be Scientists!

3 min

Say: *"Our big question is: Where does rain come from? Here is a clue — the answer starts with a puddle on the ground!"*

Watch For

Pair-share: Where do you think rain comes from? Collect answers. Then reveal: Rain is made from water that was once in a puddle, a river, or the ocean!

Slide 4: What Are the Parts?

5 min

Say: *"Our model has two parts. Sun's Heat is how warm the sun is — we cannot control the sun! Evaporation Speed is how fast water goes up into the sky — that depends on the sun's heat."*

Watch For

Use sorting mats. Students place Sun's Heat on OUTSIDE and Evaporation Speed on INSIDE. Warm hands together (that is heat!) then wave fingers up (that is evaporation!).

Slide 5: How Are They Connected? 5 min

Say: *"When the sun is REALLY hot, water goes up into the sky FAST! On a cool day, it goes slowly. More heat means faster evaporation — that is POSITIVE!"*

Watch For

Demonstrate: place a wet paper towel near a warm lamp and another far away. Which dries faster? The one near the heat! Students give thumbs up for positive connection.

Slide 6: What Did We Find Out? 5 min

Say: *"We discovered the water cycle! The sun heats water, it floats up as gas, makes clouds, and falls back as rain. It goes around and around forever!"*

Watch For

Act out the water cycle as a class: crouch down (puddle), slowly rise (evaporation), gather together (cloud), fall down (rain). Repeat! Complete sentence frame.

Slide 7: STEM Challenge 10 min

Say: *"Now let us BUILD a water cycle! We will put water in a bag, tape it to a sunny window, and watch the water travel just like real rain!"*

Watch For

Distribute zip-lock bags, water, blue food coloring. Students add water, seal bags, and draw sun/cloud/rain arrows on them with markers. Tape to sunny window. Check throughout the day for condensation drops.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Sun's Heat
- **INTERNAL:** Evaporation Speed

Relationships

- **Sun's Heat to Evaporation Speed:** POSITIVE (+) — When the sun is hotter, water heats up more and turns into gas faster — evaporation speeds up because more heat energy makes water molecules move faster.

Key Discoveries

- When the sun heats water, it turns into invisible gas and floats up into the sky — that is evaporation
- The water collects in the sky and makes clouds, and when clouds get too heavy, it rains
- Water goes around and around in a cycle — puddle to sky to cloud to rain, over and over!

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Rain is new water that the clouds make"	Clouds do not create new water. Rain is made from water that evaporated from the ground, rivers, and oceans. The same water has been recycling through the water cycle for billions of years — the water you drink today might once have been in a dinosaur's lake!

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Water disappears when puddles dry up"	Water never disappears — it changes form. When a puddle dries up, the liquid water turned into water vapor (invisible gas) and floated into the air. The water is still there; you just cannot see it because it turned into a gas.